
US-D1 Pro Radar Altimeter User Manual

**D00.00.06
August 12, 2026**

Revision History

Version Number	Date	Notes
D00.00.01	Mar 28, 2025	Initial
D00.00.02	August 18, 2025	Update for firmware v1.0 release
D00.00.03	December 8, 2025	Update for firmware v1.5 release
D00.00.04	April 22, 2026	Update for firmware v1.17 release
D00.00.05	July 20, 2026	Update for firmware v1.26 release
D00.00.06	August 12, 2026	Update Section 8 with Mission Planner update instructions

IMPORTANT NOTICE

This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) This device must accept any interference received, including interference that may cause undesired operation.

This equipment has been tested and found to comply with the limits for a Class A device (commercial or industrial use), pursuant to Part 15 of FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communication. Operation of this equipment in a residential area is likely to cause harmful interference, in which case the user will be required to correct the interference at his/her own expense.

NOTE:

- Only peripherals complying with FCC limits may be attached to this equipment. Operation with non-compliant peripherals or peripherals not recommended by Einstein.
- Modification of this device may void the user's authority to operate the equipment
- Maintain 20cm of separation for the device during operation.

IMPORTANT NOTICE
AVIS IMPORTANT

ISED Radiation Exposure Statement:

This equipment complies with ISED RF radiation exposure limits set forth for an uncontrolled environment. This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter. This equipment should be installed and operated with a minimum distance of 20 cm between the radiator & your body.

IC exposition aux radiations:

Cet équipement est conforme aux limites d'exposition aux rayonnements RF de l'isedc établies pour un environnement non contrôlé.

Cet émetteur ne doit pas être co-localisés ou fonctionner en conjonction avec une autre antenne ou émetteur. Cet équipement doit être installé et utilisé avec une distance minimale de 20cm entre le radiateur & votre corps.

Regulatory Approval Logos



IC: 26683-USD1PROV1

1. Product Description

1.1. US-D1 Pro

The US-D1 Pro is the second generation of **short-mid range Radar Altimeter**. It uses the principles of radio detection and ranging to determine the altitude of the vehicle. A microwave signal is transmitted out of the sensor, reflects off the target, and is received by the sensor. Distance from the sensor to the terrain (altitude) is derived from the difference in time from when the signal is sent from the sensor to when the signal is received by the sensor.

1.2. Compliances

1.2.1 Radio Equipment, EMC/EMI, RF

This equipment is designed to comply with the limits for a Class A device (commercial or industrial use), pursuant to Part 15 of the FCC rules or RED of 2014/53/EU. Please see the previous pages for more information or consult Ainstein.

1.2.2 Radio Technical Commission for Aeronautics (RTCA)

This equipment is designed to comply with RTCA DO-160, 178, and 254.

1. Environmental (DO-160) - This equipment is designed to comply with the RTCA DO-160G. However, we have not performed the official test and have not obtained the certification yet.
2. Software (DO-178) - This equipment is designed to comply with the RTCA DO-178C DAL C. However, we have not performed the official test and have not obtained the certification yet.

Electronic Hardware (DO-254) - This equipment is designed to comply with the RTCA DO-254 DAL C. However, we have not performed the official test and have not obtained the certification yet.

2. Installation Guidelines

Warning! Follow the manufacturer's regulations when mounting, modifying, repairing, and maintaining equipment. The manufacturer assumes no responsibility for any accidents caused by incorrectly mounted or incorrectly maintained equipment. The manufacturer assumes no responsibility for the system being incorrectly applied or the system being programmed in a manner that jeopardizes safety.

The following instructions are critical to the proper operation of the US-D1 Pro device. Failure to install the unit according to the given instructions can result in malfunction of the device.

2.1. Mounting Angle and Position

When mounting the device, the front face of the US-D1 Pro beam must be **perpendicular** to the ground below the vehicle. There should not be any angle of inclination in any direction while the device is fastened to the vehicle.

In addition to the mounting angle, it is recommended that the angle between the radar beam and other obstacles (e.g., landing gear) is greater than 50° and material is a low-permittivity type to achieve optimal performance.

2.2 Mounting to a vehicle

The device should be secured to the vehicle and point to the desired targets, where it is not free to move in any direction. For example, if the desired target is the ground, the radar need mount on the bottom of the vehicle and facing down.

2.3. Line of Sight Clearance

Keep the face of the radar clean, and do not cover it with any additional materials. Any coatings, coverings, and modifications to the radome can degrade the performance of the radar device.

Additionally, keep any unexpected objects out of the radar's FoV (Field of View). Obstructions to the US-D1 Pro's field of view will cause a decrease in the performance of the radar. It is highly recommended that the angle between the radar beam and expected objects (e.g., landing gear) be greater than 50° and material be a **low-permittivity** type to achieve optimal performance.

2.4. Integration Requirements

The US-D1 Pro outputs altitude (AGL) measurements, vertical speed measurements, signal-to-noise ratio (SNR), and Measurement Confidence when operational. When integrating the US-D1 Pro radar altimeter, it is recommended to use Measurement Confidence to properly filter out erroneous altitude values.

A filtering algorithm should be used to estimate vehicle position, velocity, and angular orientation based on rate gyroscopes, accelerometer, compass, GPS, airspeed, and barometric pressure measurements in addition to the recorded US-D1 Pro measurements. Sensor redundancy is strongly advised for the US-D1 Pro.

3. Operational Requirements

The US-D1 Pro will perform optimally if the operational requirements below are fully satisfied. Failure to meet the operational requirements may cause a decrease in performance, accuracy, or reliability of the US-D1 Pro altimeter and is not advised.

3.1 Obstruction to the US-D1 Pro Field of View

Objects or vehicle structures that are located within the US-D1 Pro's conical field of view which obstruct the radar's view of the ground may cause multipath reflections or other degradative phenomena to occur. The US-D1 Pro should be mounted a safe distance away from the landing gear and any other components of the vehicle below it.

3.2 Excessive Pitch/Roll

Pitch and roll angles that exceed 20° and 20° respectively may also cause the performance of the US-D1 Pro to worsen. This sensitivity increases with altitude.

3.3 Terrain

Terrain with poor reflectivity may cause the performance of the US-D1 Pro to worsen. Flying at the limit of the US-D1 Pro's range over dry, loose soil, such as tilled farmland, or sand is not recommended.

3.4 Power Source

Any power source used to operate the US-D1 Pro that does not provide the proper power and voltage can worsen the performance of the device or cause its operation to stop altogether. Moreover, if the power voltage is significantly more than the input voltage upper limit, it may permanently damage the device.

3.5 Orientation of US-D1 Pro

The US-D1 Pro must be mounted upon the vehicle in such a manner as to be fully horizontal, its radome directly facing the ground. Any angle of inclination may degrade performance.

3.6 Minimum and Maximum Operating Altitude

Operating the US-D1 Pro at altitudes below its minimum altitude and above 120 meters will result in a degradation of performance and potentially erroneous measurements.

4. Technical Specifications

Table 1: Specifications

Operating Altitude (Minimum ~ Maximum)	The Minimum Range~120m ¹
Altitude Accuracy	±0.1m(@≤10m); 1% (@>10m)
Field of view (FOV)	Azimuth ±20° ² Elevation ±20°
Frequency Band	24 GHz ISM Band
Transmit EIRP	Typ. 2.4 dBm ³
Update Rate	20Hz
Recommended Supply Voltage	DC 12~14V
Supply Voltage Range	DC 9~16V
Harness/Connector	MOLEX-349671001
Power consumption	Typ. 3.5W ⁴
Temperature Range	-40°C~85°C (Operational) -55°C~85°C (Storage)
Weight	140 g
Size	101 mm * 76mm * 24.71 mm
Interface	Two CAN-FD channels 1st channel for Private CAN messages and OTA upgrade ⁵ 2nd channel for Public CAN messages and features ⁶
OTA Upgrade Method	- Private CAN by Einstein's GUI - Public CAN by tools that support it (Public CAN GUI, Mission Planner, etc)
Protection level	Contact Einstein for Details ⁷
Time Synchronization	- Public CAN - Supported in the node status reporting - Private CAN - NOT supported

Notes:

1. This is the recommended AGL (Above Ground Level) altitude range.
 - a. The maximum altitude cap is 150m.
 - b. The minimum range depends on the actual mounting height and the radar's minimum detection.
 - i. For fixed-wing aircraft and similar vehicle types, this minimum range is set as 0.7 ~ 2 meters.

- ii. For Copter aircraft and similar vehicle types, this minimum range is set as 0.5 ~ 2 meters.
2. FOV defined here is the same as the antenna pattern specs
3. This EIRP value is the measured transmitter PEAK power from US-D1 Pro at 24GHz fundamental band, including the Transmitting Antenna Gain.
4. This is the mean power consumption in steady operation. The peak power consumption might be greater than it.
5. OTA upgrade via Private CAN is performed by the Ainstein GUI via a wired method
6. Supported Public CAN feature and functionality includes, but is not limited to (See details in Section 6.2):
 - a. Export Node Status message
 - b. Export Sensor Measurement message
 - c. Time synchronization
 - d. Node parameter configuration
 - e. Firmware update
7. For the details of Ingress Protection, please contact Ainstein. The Protection Level is only for the US-D1 Pro device with a proper plug; it may not include the cable.

5. Communication Protocol For US-D1 Pro

US-D1 Pro has two channels of CAN interface in hardware, and it does support two CAN communication protocols: Private CAN and Public CAN.

5.1 Private CAN

5.1.1 Private CAN Specification

- CAN Interface Type - Default of CAN 2.0
- Baud Rate - Default of 1 Mb/s in CAN 2.0
- Support Frame Format and ID - Default of Extended Frame with 0x90002
- Standard: CAN FD (maintains backward compatibility with CAN 2.0), ISO 11898-1

The type of CAN frame used and CAN ID of the device can be customized. The CAN frame Data Field is defined in the table below.

From	US-D1 Altimeter	To	Receiver
Byte	Data	Note	

1	0x**	Altitude (Most Significant 8 Bits) ¹
2	0x**	Altitude (Least Significant 8 Bits)
3	0x**	SNR(Most Significant 8 Bits) ²
4	0x**	SNR(Least Significant 8 Bits)
5	0x**	Confidence ³
6	0x**	Reserved ⁴
7	0x** (4-bit)	Counter ⁵
8	(Reserved)	(Reserved)

Table 5.1: Private CAN Data Format Definition

Notes:

1. Altitude: It is the AGL altitude reading. Unit: centimeters (0.01 m); Unsigned value;
2. SNR: It is the Signal-to-Noise Ratio for the reflection signal. Unit: dB; Unsigned value; Note: there is no Absolute Threshold in SNR for users' reference
3. Confidence: It indicates the radar's measurement confidence in percentage; its range is from 0 to 100. Unit: %.
 - a. Absolute threshold: 0 - Invalid detection
 - b. Other: 1 ~ 100 - Reference for users
4. This Byte is reserved as 0.
5. Counter: It indicates the CAN message's rolling counter and automatically increments by 1 for a new CAN message. Range from 0 to 15;
6. There are two cases that indicate the 'Invalid' reading
 - a. When the device is out of range (greater than the maximum altitude cap or smaller than the minimum range configuration).
 - b. If there is no valid measurement.

Note: both 'Invalid' reading cases mean the US-D1 Pro is still operational with invalid measurement data export

5.1.2 Private CAN Configuration Options

No.	Parameter	Configurable Range	Default
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1	CAN Interface Type	• CAN 2.0 • CAN FD	CAN 2.0
2	Frame Format	• Standard Frame • Extended Frame	Extended Frame
3	ID	• Standard Frame: 0x000 ~ 0x7FF; • Extended Frame: 0x00000000 ~ 0x1FFFFFFF	0x90002
4	Baud Rate	• CAN 2.0: ◦ 500 Kbps ◦ 1000 Kbps • CAN FD: ◦ 500 Kbps ◦ 1000 Kbps	1000 Kbps
5	Bit Rate Switch	• CAN FD: ◦ 500 Kbps ◦ 1000 Kbps ◦ 2000 Kbps ◦ 5000 Kbps	1000 Kbps

Table 5.2: Private CAN Configurable Parameter

5.2 Public CAN Specification

US-D1 Pro Public CAN uses the primary CAN protocol used by the ArduPilot and PX4 projects for communication with CAN peripherals. It is an open protocol with open communication, specification, and multiple open implementations. For details of this Public CAN, please refer to the official website.

- CAN Interface Type: CAN 2.0
- Baud Rate – Default of 1 Mb/s
- CAN Standard: CAN 2.0B
- Transfer Priority – Default of 24
- Source Node ID – Default ID 32
- Node Status Message
 - Data type ID: 341
 - Broadcasting at 1 Hz
- Range Sensor Measurement

- Data type ID: 1050
- Broadcasting at 20 Hz
- Sensor ID
 - Default of 1
 - Configuration Range: 0-255

Compatible with Ardupilot and PX4 Public CAN driver. Follow the official instructions to conduct the autopilot configuration.

5.2.1 Public CAN Node Status

US-D1 Pro Public CAN node status spec	Public CAN message field	Setting	
Time Synchronization	uptime_sec	1 second	
Node Health	health		
		HEALTH_OK	Radar in normal operation
		HEALTH_WARNING	Radar in normal operation. Users shall use it with caution.
		HEALTH_ERROR	Radar in stress operation. Users shall stop using it.
		HEALTH_CRITICAL	Radar failed to report. Users shall stop using it.
Current Node Status	mode		
		MODE_OPERATIONAL	Radar runs in Application firmware and is in operation. It reports detection data

			and messages.
		MODE_INITIALIZATION	Radar is booting; no data reporting.
		MODE_MAINTENANCE	Radar runs in Bootloader firmware and reports no detection data and messages.
		MODE_SOFTWARE_UPDATE	Radar is under Firmware update
		MODE_OFFLINE	N.A.
Sub mode - Not used currently	sub_mode	N.A. Fixed in 0	
Optional, vendor-specific node status code	vendor_specific_status_code	N.A. Fixed in 0	

Table 5.3: Public CAN (Public CAN) Node Status Data Format Definition

5.2.2 Public CAN Sensor Measurement

US-D1 Pro Public CAN sensor measurement spec	Public CAN message field	Setting
Timestamp for synchronization	timestamp	US-D1 Pro is a Time Slave node; it synchronizes the timestamp with the Time Master node on the bus. If there is no master clock node, it reports 0
Sensor ID	sensor_id	Default is 1; Configurable
Beam Orientation	beam_orientation_in_body_	N.A. Fixed at 0

	frame	
Field of View	field_of_view	US-D1 Pro ' <u>Confidence</u> ' data is filled in this field
Radar's Sensor Type	sensor_type	3 – Radar
Radar's Reading Type	reading_type	0 – N.A. in US-D1 Pro 1 – When Range reading is <= 120 m, including 0 reading 2 – N.A. in US-D1 Pro 3 – When Range reading is > 120 m
Range	Range	US-D1 Pro range data in <u>meters</u>

Table 5.4: Public CAN (Public CAN) Sensor Measurement Data Format Definition

5.2.3 Public CAN Configuration Option

No.	Parameter	Configurable Range	Default
1	Transfer Priority	0~31	24
2	Node ID	0~127 0 - Reserved for DNA 126, 127 - Reserved for debug	32
3	Sensor ID	0~255	1
4	CAN Baudrate	- CAN 2.0: 500 Kbps 1000 Kbps	1000 Kbps
5	Setting Height (Minimum)	- Copter: 0.5~ 2 - Fixed Wing: 0.7 ~ 2	0.5 0.7
6	Flight Mode	- Copter: 1 - Fixed Wing: 2	1

Table 5.5: Public CAN Configurable Parameter

NOTE:

- When connecting the US-D1 Pro to ArduPilot, users must ensure that the RNGFNDX_ADDR parameter matches the sensor ID of the radar (i.e., if Sensor ID = 1, then RNGFNDX_ADDR = 1)

6. Hardware Interface

US-D1 Pro's connector and its receptacle are automotive-grade and have the IP69K rating.

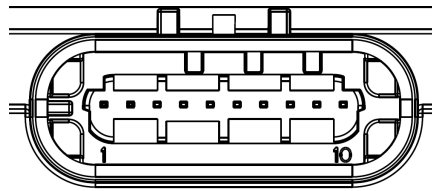


Figure 5.1: US-D1 Pro connector

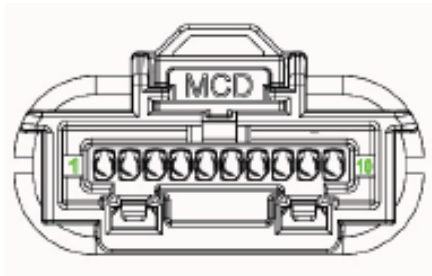


Figure 6.2: US-D1 Pro Connector's Receptacle - MOLEX-349671001

Pin	Definition	Function	Remark
1	CANL_PRI	Private CAN-L low	For Private CAN channel, an external 120Ω resistor is required
2	CANH_PRI	Private CAN-H high	
3	CANL_PUB	Public CAN-L low	For the Public CAN channel; 120Ω resistor requirement is up to the bus setting
4	CANH_PUB	Public CAN-H high	
5	NC	none	
6	NC	none	
7	NC	none	
8	GND_IN	Power ground	The power ground and signal ground are short-circuited and

			isolated from the shell (to improve ESD anti-interference ability)
9	NC	none	
10	DC_IN	Power input	9V~16V DC Input

Table 6.1: Pinout Definition

7. Mechanical Specifications

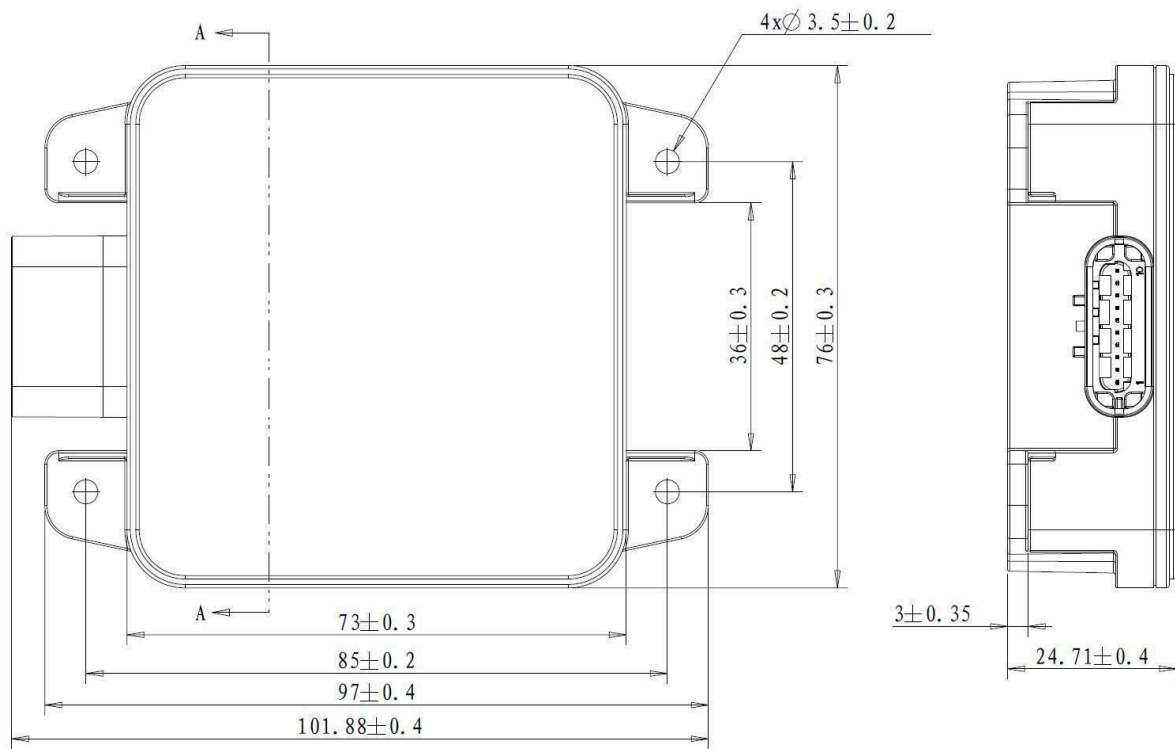


Figure 7.1: US-D1 Pro Dimension (Unit: mm)

8. Firmware and Parameter Update

NOTICE:

- Ainstein recommends that users follow either section 8.1 or 8.2 to perform the firmware and parameter update by either the Private CAN interface or the Public CAN interface.
- **THE CURRENT US-D1 PRO APPLICATION FIRMWARE IS VERSION 1.26, AND THE BOOTLOADER IS VERSION 1.5.**

8.1 Private CAN Firmware and Parameter Update

IMPORTANT: THE US-D1 PRO BOOTLOADER FIRMWARE MUST BE UPDATED TO ITS LATEST VERSION BEFORE UPDATING THE APPLICATION FIRMWARE

To perform the Application firmware and parameter update via Private CAN, Ainstein will provide the following tool and file.

1. US-D1 Pro OTA Updater GUI Tool
2. Firmware Flash File for Bootloader Firmware and Application Firmware (.bin)
3. Data Identifier (DID) sheet for Parameter Configuration

Users should have the following tools.

1. PCAN-FD to USB Adapter
2. PC in Windows OS

Note:

- Important: When using the “US-D1 Pro OTA Updater GUI Tool” to perform Bootloader firmware, Application firmware, and Data Identifier (DID) sheet file updates, please follow the exact sequence and steps in Section 8.1.1 and 8.1.2. The CAN transceiver in US-D1 Pro hardware is CAN-FD, and it is compatible and can be configured to CAN 2.0.
 - The interface between this US-D1 Pro OTA Updater GUI Tool and US-D1 Pro is fixed in CAN-FD, instead of CAN2.0. Therefore, a **PCAN-FD to USB Adapter is required**.

8.1.1 ‘US-D1Pro OTA Updater’ and Firmware Update Instruction

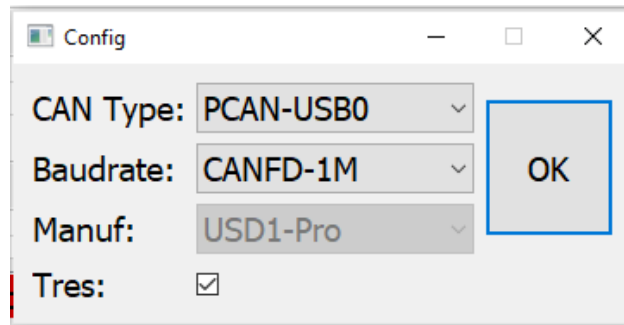
Note: In the following example, the actual parameter values may not be the same as the latest firmware version.

The US-D1 Pro OTA Updater is the GUI tool to perform the Bootloader firmware and Application firmware update via Private CAN. The following steps instruct users how

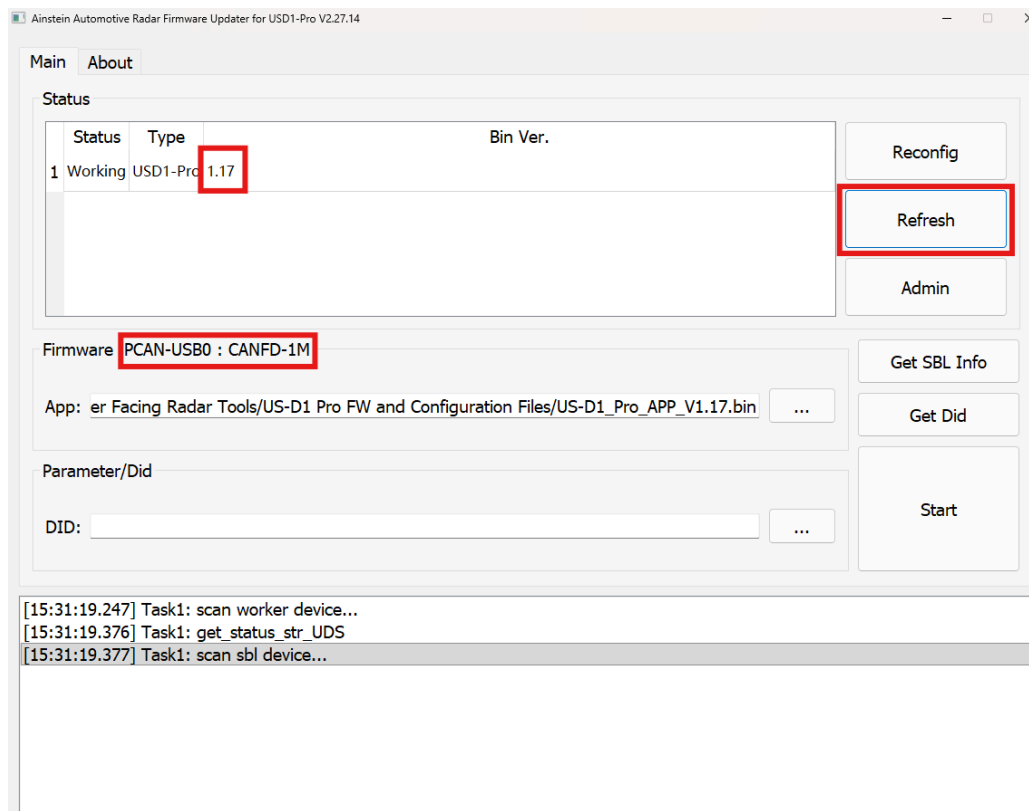
to update either only the Application Firmware (Section 8.1.1.2) or both the Bootloader and Application Firmware (Section 8.1.1.3):

8.1.1.1 Check Application and Bootloader Firmware Versions

1. Open the tool and enter the tool configuration window
2. Select the CAN Type as "PCAN-USB0"; Baudrate as "CANFD-1M"; Then, click 'OK' to enter the main window



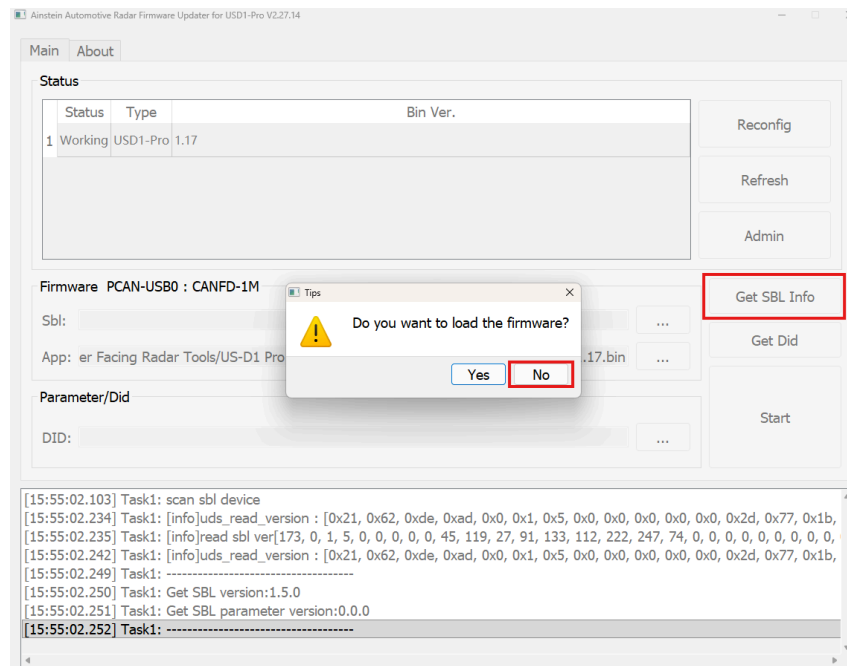
3. In the main window, click 'Refresh' to inquire about US-D1 Pro's current status and Application firmware version



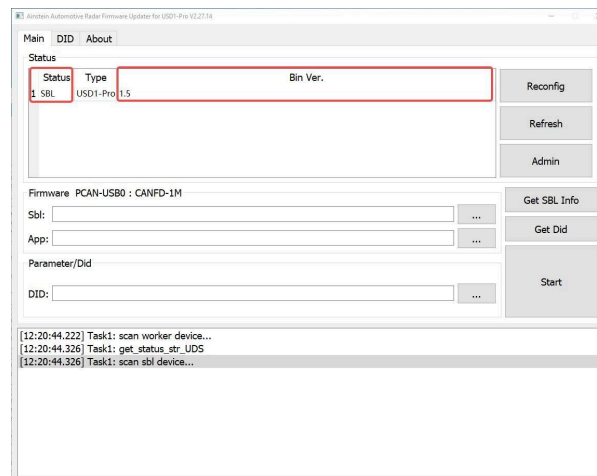
Note:

- Status - Working: US-D1 Pro is operating in the Application firmware
- Type - USD1-Pro: GUI recognize the device as US-D1 Pro correctly

- Bin Ver - 1.17.0: Current Application firmware version is v1.17.0 (or v1.17)
4. Check the Bootloader version
 - a. Click 'Get SBL Info', then click 'No' in the pop-up window



- b. Click 'Refresh'; it will show the Bootloader firmware information.
 - i. Bin Ver - 1.5: Current Bootloader firmware version is v1.5

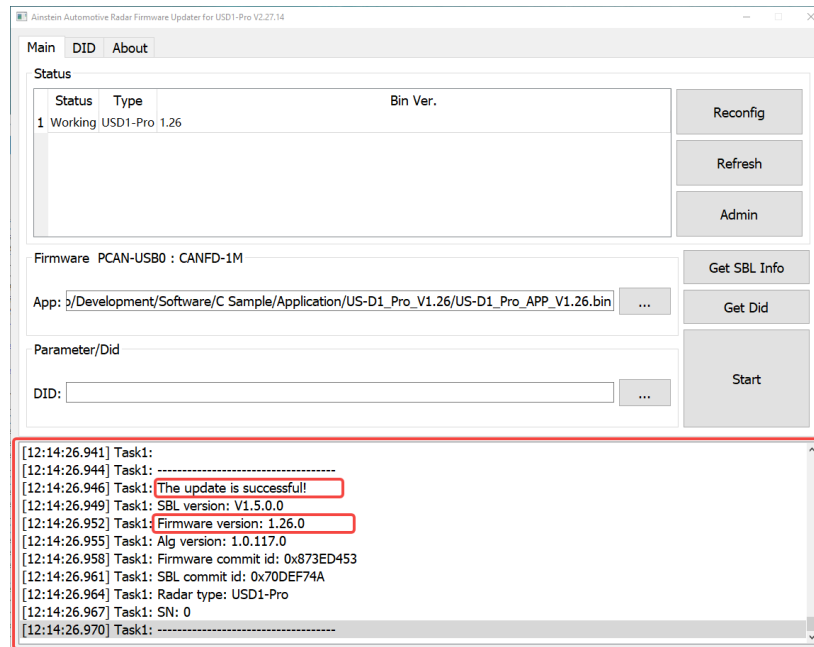


- c. Recover to Application mode: Click 'Get SBL Info', then click 'Yes' in the pop-up window to recover the device, or wait for 30 seconds, press 'Refresh', and the device will automatically recover from Bootloader mode to Application mode.

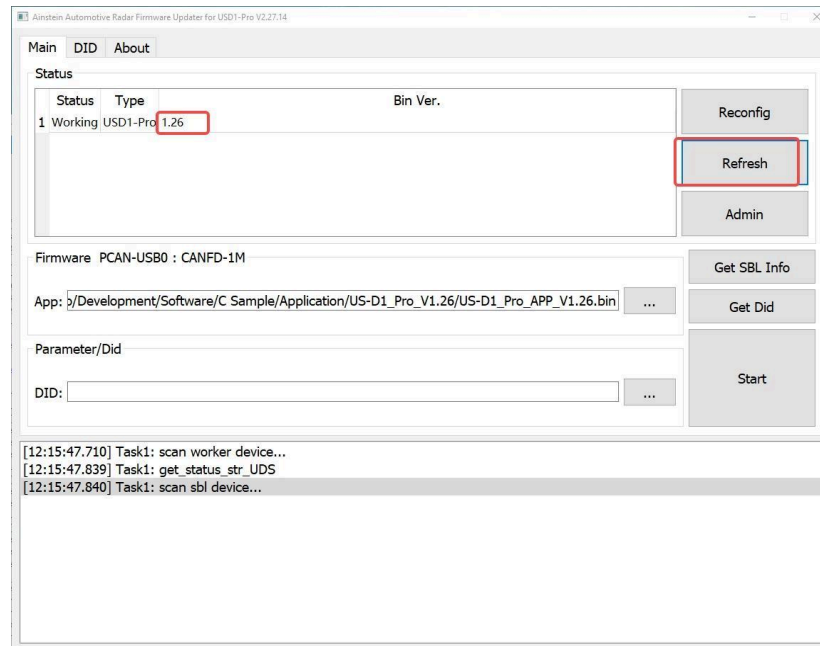
8.1.1.2 Only Flash the Application Firmware

IMPORTANT: THE US-D1 PRO BOOTLOADER FIRMWARE MUST BE UPDATED TO ITS LATEST VERSION BEFORE UPDATING THE APPLICATION FIRMWARE

1. Perform steps 1-4 in Section 8.1.1.1 to verify Application and Bootloader firmware versions
2. ENSURE ALL FILE SELECTION WINDOWS ARE EMPTY
3. Select the application firmware file in the blank 'App:' window
 - a. Click 'Start'
 - b. Wait until the Application firmware update process completes. Once it completes and succeeds, it will show the messages.



4. Click 'Refresh' to inquire about the US-D1 Pro's new Application firmware version

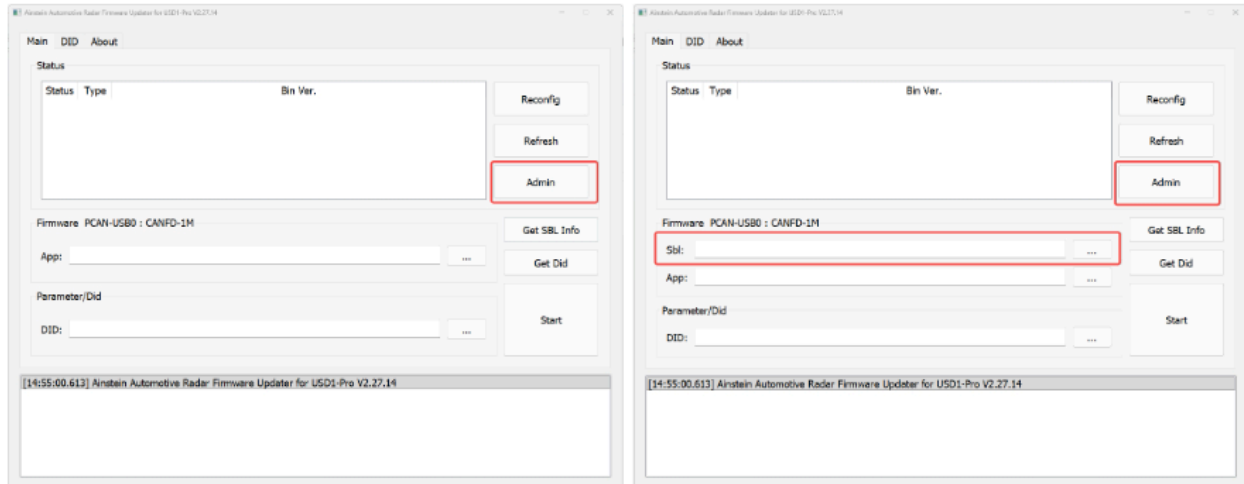


5. Power cycle the US-D1 Pro

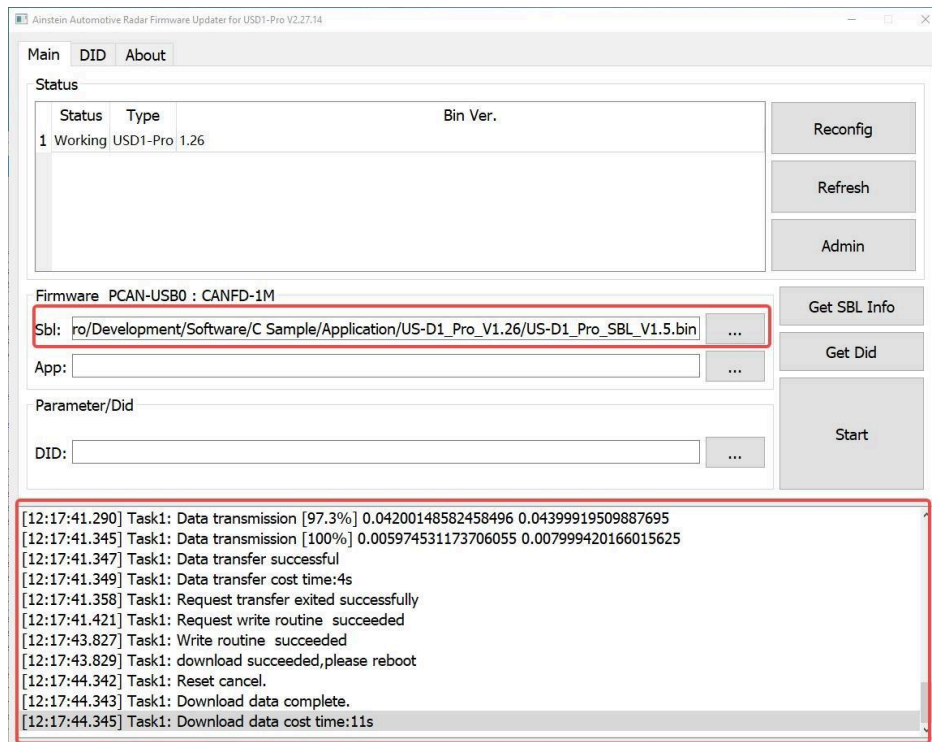
8.1.1.3 Flash both Bootloader and Application Firmware

IMPORTANT: THE US-D1 PRO BOOTLOADER FIRMWARE MUST BE UPDATED TO ITS LATEST VERSION BEFORE UPDATING THE APPLICATION FIRMWARE

1. Perform steps 1-4 in Section 8.1.1.1 to verify Application and Bootloader firmware versions
2. Click the 'Admin' button; then the SBL blank window will be shown for Bootloader firmware file selection



3. ENSURE ALL FILE SELECTION WINDOWS ARE EMPTY
4. Perform Bootloader firmware update:
 - a. Select the Bootloader firmware file
 - b. Click 'Start'
 - c. Double check to perform the Bootloader update is needed, then click 'OK' in the pop-up window.
 - d. Wait until the Bootloader firmware update process completes. Once it completes and succeeds, it will show the messages.



5. Power cycle the US-D1 Pro

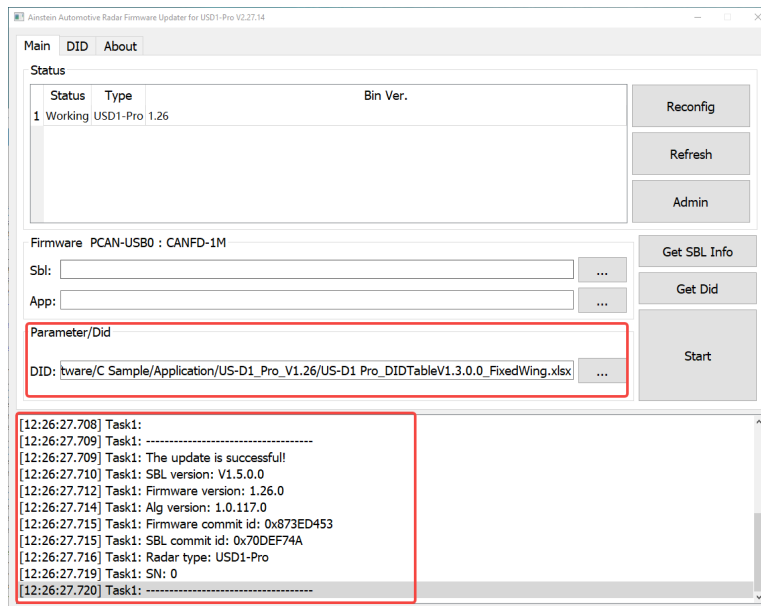
6. Perform step 4 from Section 8.1.1.1 to verify the Bootloader version
7. Follow steps 2-5 in Section 8.1.1.2 to flash the Application Firmware

8.1.2 Parameter Configuration Instruction

Under the private CAN protocol, the configurable parameter information can be found in Section 5.1.2. To perform the parameter configuration, simply use the same GUI tool and follow the steps below.

8.1.2.1 - Write DID file to configure parameter

1. Perform parameter update.
 - a. Perform steps 1-4 in Section 8.1.1.1
 - b. **Caution:** Clear the content in the App and/or SBL blanks, if there are any.
 - c. Select the Data Identifier (DID) sheet file
 - d. Click 'Start'
 - e. Wait until the parameter update process completes. Once it completes and succeeds, it will show the messages.



2. Power cycle the US-D1 Pro

8.1.2.2 - Read DID file to verify parameter

1. Recover the US-D1 Pro to Application mode
2. Click 'Get Did'

3. The key parameters will be shown in the bottom window
 - a. Minimum Height (Radar Height Above Ground)
 - b. Flight Mode

8.1.3 Data Identifier (DID) sheet

In the DID sheet, the mapping between the values in cells and configurable parameters can be found below.

Parameters	Data Type	Length	Read-Write Property	Session Type	Notes	Value									
crcCode	uint32_t	1	R	E		1									
DidTableLength	uint32_t	1	R	E		1	1	2					4	5	
Did_D028	uint8_t	8	RW	E	CAN Parameter Settings	0x1	0x1	0x0	0x0	0x3	0xE8	0x3	0xE8		
Did_D029	uint32_t	1	RW	E	Private CAN ID	0x00090002									
Did_D030	float	1	RW	E	Radar Height Above Ground	0.7									
Did_D031	uint32_t	1	RW	E	Flight Mode	0x2									

Note:

1. Only the above 7 cells are supported in the US-D1 Pro parameter configuration in the DID table v1.3.0.0
2. Numbers 1, 2, 4, and 5 have a drop-down menu; users can select it from there.
3. No. 3 is the Private CAN ID; please use the valid extended CAN ID format filled in this cell (e.g., 0x00090002).
4. No. 6 is the Radar Height Above Ground; please type the number with one decimal place (e.g., 0.5) from 0.5 to 2.0 in this cell.

IMPORTANT: The minimum setting height for the fixed-wing/VTOL parameter is 0.7 meters, whereas the minimum setting height for the multi-rotor parameter is 0.5 meters.

5. For CAN 2.0, select the same value for Cells 4 and 5; for CAN FD, they can be either the same value or different.
6. No. 7 is the vehicle type selection. 1 is the Multi-Rotor vehicle; 2 is the Fixed-Wing/VTOL vehicle; any other value is prohibited.
 - a. If the take-off and landing vertical speed is faster than $\pm 6.9\text{m/s}$, please select the Fixed-Wing mode.

8.2 Public CAN Firmware and Parameter Update

As Public CAN is the public communication protocol in the industry, Ainstein recommends users to use DroneCAN GUI Tool (cross-platform free open source application for Public CAN bus management and diagnostics) or Mission Planner (versatile ground control station (GCS) software developed by ArduPilot) to perform the firmware and parameter update. Refer to the official instructions for details of the steps.

Please see the configurable parameter in Section 5.2.3, and contact Ainstein for more assistance.

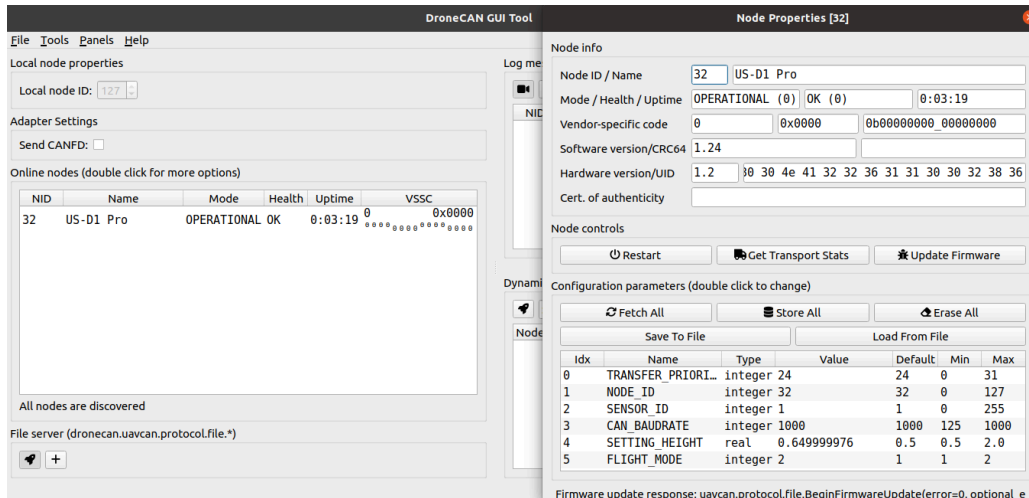
8.2.1 DroneCAN GUI Tool Firmware Update Instructions

Unlike Private CAN, there is no designated sequence required with Public CAN for updating the bootloader or application firmware.

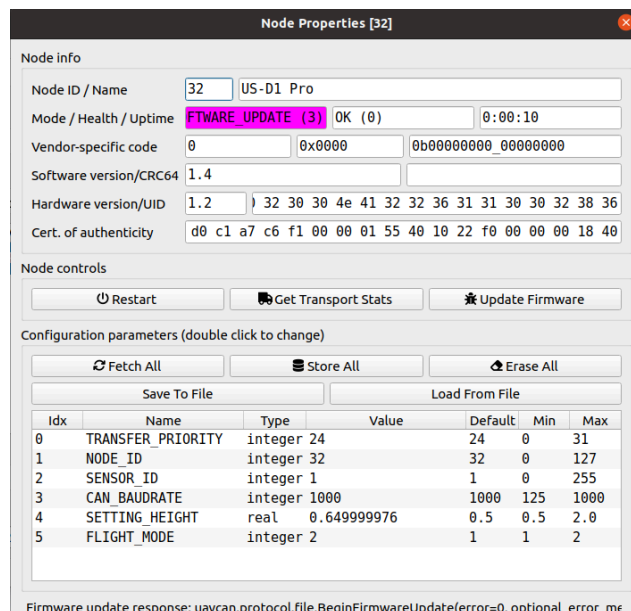
Here is an example of using the DroneCAN GUI tool to perform the firmware update.

Note: In the example, the actual parameter value may not be exactly the same as the latest firmware version.

1. Launch the Public CAN GUI tool
2. Click US-D1 Pro Node and check all information



3. Perform the firmware update
 - a. Click 'Update Firmware'
 - b. Select a firmware file (.bin), Bootloader, or Application. Here, the Application firmware file is used as an example.
 - c. Wait until the firmware update process completes. GUI shows 'SOFTWARE UPDATE'.



4. Once it completes and succeeds, it will automatically reboot and operate.
 - a. After rebooting, the Mode will show 'MAINTENANCE' when the radar is in Bootloader.

The screenshot shows the 'Node Properties [32]' window. The 'Node info' section displays the following data:

- Node ID / Name: 32 US-D1 Pro
- Mode / Health / Uptime: MAINTENANCE (2) OK (0) 0:00:00
- Vendor-specific code: 0 0x0000 0b00000000_00000000
- Software version/CRC64: 1.4
- Hardware version/UID: 1.2 32 30 30 4e 41 32 32 36 31 31 30 30 32 38 36
- Cert. of authenticity: d0 c1 a7 c6 f1 00 00 01 55 40 10 22 f0 00 00 00 18 40

The 'Node controls' section includes buttons for Restart, Get Transport Stats, and Update Firmware. The 'Configuration parameters (double click to change)' section includes buttons for Fetch All, Store All, Erase All, Save To File, and Load From File. Below these buttons is a table of configuration parameters:

Idx	Name	Type	Value	Default	Min	Max
0	TRANSFER_PRIORITY	integer 24		24	0	31
1	NODE_ID	integer 32		32	0	127
2	SENSOR_ID	integer 1		1	0	255
3	CAN_BAUDRATE	integer 1000		1000	125	1000
4	SETTING_HEIGHT	real	0.649999976	0.5	0.5	2.0
5	FLIGHT_MODE	integer 2		1	1	2

The status bar at the bottom shows: Firmware update response: uavcan.protocol.file.BeginFirmwareUpdate(error=0, optional_error_me

- b. After the radar switches to Application, the Mode shows 'OPERATIONAL'. The software version shows the updated Application firmware version of 1.24.

The screenshot shows the 'Node Properties [32]' window. The 'Node info' section displays the following data:

- Node ID / Name: 32 US-D1 Pro
- Mode / Health / Uptime: OPERATIONAL (0) OK (0) 0:17:29
- Vendor-specific code: 0 0x0000 0b00000000_00000000
- Software version/CRC64: 1.24
- Hardware version/UID: 1.2 32 30 30 4e 41 32 32 36 31 31 30 30 32 38 36
- Cert. of authenticity:

The 'Node controls' section includes buttons for Restart, Get Transport Stats, and Update Firmware. The 'Configuration parameters (double click to change)' section includes buttons for Fetch All, Store All, Erase All, Save To File, and Load From File. Below these buttons is a table of configuration parameters:

Idx	Name	Type	Value	Default	Min	Max
-----	------	------	-------	---------	-----	-----

5. The same process can be conducted for the bootloader firmware update.

8.2.2 DroneCAN GUI tool Parameter Fetch and Configuration

Here is an example of using the Public CAN GUI tool to perform the parameter fetch and configuration under firmware v1.24.

Node Properties [32]

Node info

Node ID / Name32US-D1 Pro

Mode / Health / UptimeOPERATIONAL (0)OK (0)0:05:27

Vendor-specific code00x00000b00000000_00000000

Software version/CRC641.24

Hardware version/UID1.230 32 30 30 4e 41 32 32 36 31 31 30 30 32 38 36

Cert. of authenticity

Node controls

RestartGet Transport StatsUpdate Firmware

Configuration parameters (double click to change)

Fetch AllStore AllErase All

Save To File			Load From File			
Idx	Name	Type	Value	Default	Min	Max
0	TRANSFER_PRIORITY	integer	24	24	0	31
1	NODE_ID	integer	32	32	0	127
2	SENSOR_ID	integer	1	1	0	255
3	CAN_BAUDRATE	integer	1000	1000	125	1000
4	SETTING_HEIGHT	real	0.649999976	0.5	0.5	2.0
5	FLIGHT_MODE	integer	2	1	1	2

6 params fetched successfully

Note:

- Parameter Idx 0~2 are DroneCAN parameter
- Parameter Idx 3 - CAN_BAUDRATE is for the CAN baud rate in CAN 2.0. DroneCAN does NOT support CAN FD.
 - Same as Cell 4 in the DID table under Private CAN
- Parameter Idx 4 - SETTING_HEIGHT is for the Radar Height Above Ground; please type the number with one decimal (e.g. 0.5) from 0.5 to 2.0 in this cell.
 - Same as Cell 6 in the DID table under Private CAN

IMPORTANT: The minimum setting height for the fixed-wing/VTOL parameter is 0.7 meters, whereas the minimum setting height for the multi-rotor parameter is 0.5 meters.

- Parameter Idx 5 - FLIGHT_MODE is for the vehicle type selection. 1 is the Copter vehicle; 2 is the Fixed-Wing vehicle; any other value is illegal.
 - Same as Cell 7 in the DID table under Private CAN

8.2.3 Mission Planner Firmware Update Instructions

Unlike Private CAN, there is no designated sequence required with Public CAN for updating the bootloader or application firmware.

Here is an example of using Mission Planner to perform the firmware update.

Note: In the example, the actual parameter value may not be exactly the same as the latest firmware version.

1. Launch Mission Planner
2. Navigate to 'Setup' -> 'Optional Settings' -> 'DroneCAN/UAVCAN' window
3. Connect and verify that the US-D1 Pro is displayed in the window

ID	Name	Mode	Health	Uptime	HW Version	SW Version	SW CRC	Menu
127	org.missionplanner	OPERATIONAL	OK	00:01:58	0.0	1.3.83	0	Menu
10	org.ardupilot:0	OPERATIONAL	OK	00:03:44	1.0	1.0.0	0	Menu
41	US-D1 Pro	OPERATIONAL	OK	00:02:58	1.2	1.26.873ED4...	0	Menu

4. Perform the firmware update
 - a. Navigate to the 'Menu' button on the 'US-D1 Pro' row
 - b. Select 'Update'

ID	Name	Mode	Health	Uptime	HW Version	SW Version	SW CRC	Menu
127	org.missionplanner	OPERATIONAL	OK	00:06:06	0.0	1.3.83	0	Menu
10	org.ardupilot:0	OPERATIONAL	OK	00:07:52	1.0	1.0.0	0	Menu
41	US-D1 Pro	OPERATIONAL	OK	00:07:06	1.2	1.26.873ED4...	0	Menu

- c. When asked 'Do you want to search the internet for an update?', select No
 - d. Select a firmware file (.bin), Bootloader, or Application. Here, the Application firmware file is used as an example.
 - e. During the firmware update, the bootloader version will appear in the window under the 'SW Version' column and will remain for a short period of time after the update is completed
 - i. During that time is when users can validate that the correct bootloader has been applied

	ID	Name	Mode	Health	Uptime	HW Version	SW Version	SW CRC	Menu
	127	org.missionplanner	OPERATIONAL	OK	00:08:25	0.0	1.3.83	0	Menu
	10	org.ardupilot:0	OPERATIONAL	OK	00:10:10	1.0	1.0.0	0	Menu
▶	41	US-D1 Pro	SOFTWARE_...	OK	00:00:32	1.2	1.5.70DEF74A	01022FA4010DEC4	Menu

Progress

fw.bin

Cancel

- f. After the bootloader version displays, the application firmware will appear under the 'SW Version' column

	ID	Name	Mode	Health	Uptime	HW Version	SW Version	SW CRC	Menu
	127	org.missionplanner	OPERATIONAL	OK	00:11:34	0.0	1.3.83	0	Menu
	10	org.ardupilot:0	OPERATIONAL	OK	00:13:20	1.0	1.0.0	0	Menu
▶	41	US-D1 Pro	OPERATIONAL	OK	00:02:43	1.2	1.26.873ED4...	0	Menu

8.2.4 Mission Planner Parameter Fetch and Configuration

Here is an example of using Mission Planner to perform the parameter fetch and configuration.

1. Launch Mission Planner
2. Navigate to 'Setup' -> 'Optional Settings' -> 'DroneCAN/UAVCAN' window
3. Connect and verify that the US-D1 Pro is displayed in the window

MAVLinkCAN1
Disconnect
Filter
Inspector
Stats
After enabling SLCAN, you will no longer be able to connect via MAVLINK. You must leave this screen and wait 2 seconds before connecting again

	ID	Name	Mode	Health	Uptime	HW Version	SW Version	SW CRC	Menu
	127	org.missionplanner	OPERATIONAL	OK	00:01:58	0.0	1.3.83	0	Menu
	10	org.ardupilot:0	OPERATIONAL	OK	00:03:44	1.0	1.0.0	0	Menu
▶	41	US-D1 Pro	OPERATIONAL	OK	00:02:58	1.2	1.26.873ED4...	0	Menu

4. Perform the parameter fetch and configuration
 - a. Navigate to the 'Menu' button on the 'US-D1 Pro' row
 - b. Select 'Parameters'

	ID	Name	Mode	Health	Uptime	HW Version	SW Version	SW CRC	Menu
	127	org.missionplanner	OPERATIONAL	OK	00:16:21	0.0	1.3.83	0	Menu
	10	org.ardupilot:0	OPERATIONAL	OK	00:18:07	1.0	1.0.0	0	Menu
▷	41	US-D1 Pro	OPERATIONAL	OK	00:07:30	1.2	1.26.873ED4...	0	Menu

Parameters
Restart
Update
Update Beta
CANPassThrough Here3
CANPassThrough Here3+/4

- In the parameter window, users can modify the DroneCAN parameters, CAN baud rate, flight mode, and setting height.
- Modify the desired parameters and select 'Write Params.' Here, the fixed-wing/VTOL flight mode is used as an example

IMPORTANT: The minimum setting height for the fixed-wing/VTOL parameter is 0.7 meters, whereas the minimum setting height for the multi-rotor parameter is 0.5 meters.

Command	Δ	Value	Min	Max	Default	Fav	
CAN_BAUDRATE		1000	125	1000	1000	☐	Load from file
FLIGHT_MODE		2	1	2	1	☐	Save to file
NODE_ID		41	0	127	32	☐	Write Params
SENSOR_ID		1	0	255	1	☐	Refresh Params
SETTING_HEIGHT		0.7	0.5	2	0.5	☐	Commit Params
TRANSFER_PRIORITY		24	0	31	24	☐	All Units are in raw format with no scaling

- After writing the parameters, users can validate the correct parameters via steps 1-5